Crime Management System - UML & Use Case Diagrams

Version: 1.0.0
Generated: 2025

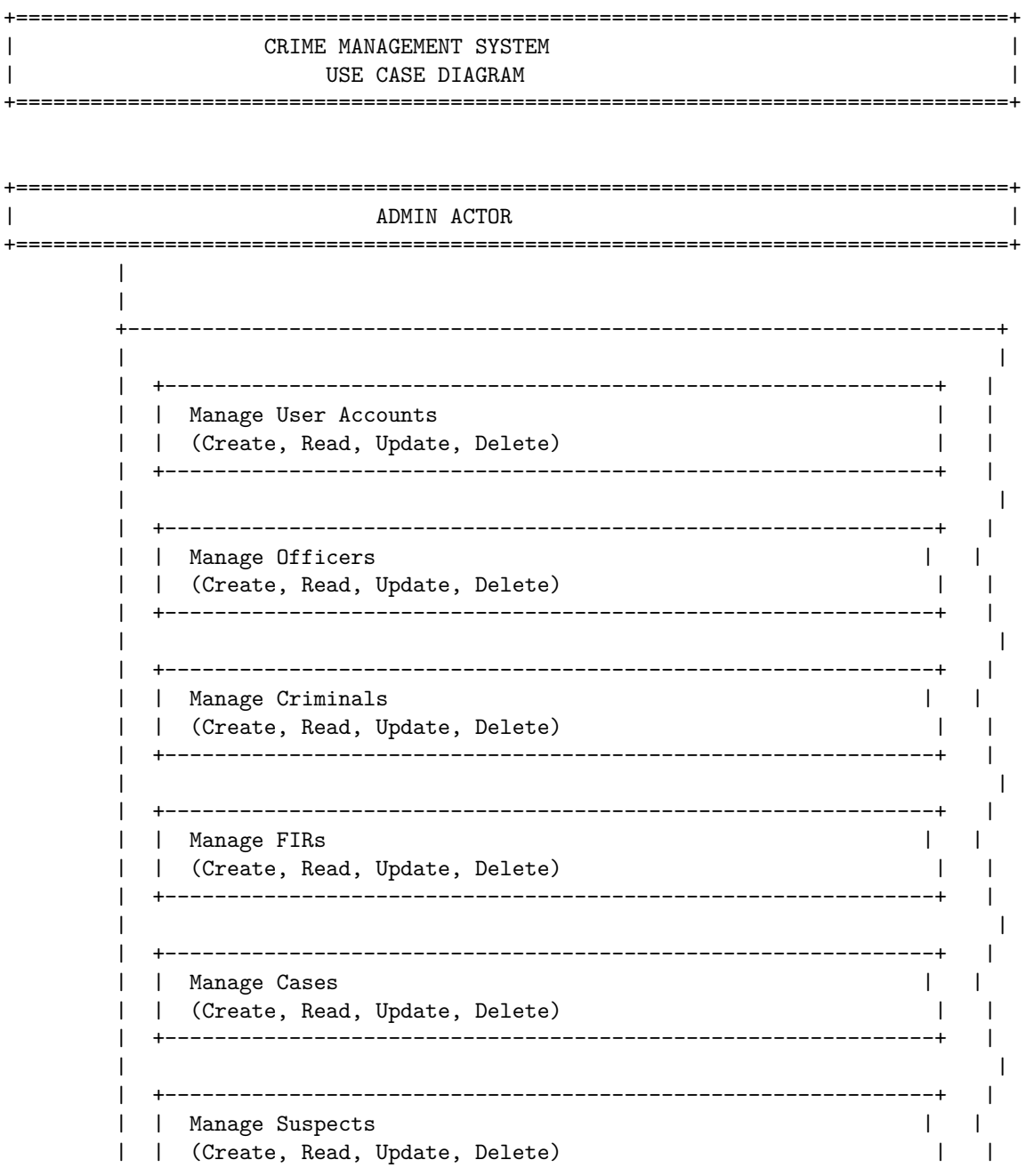
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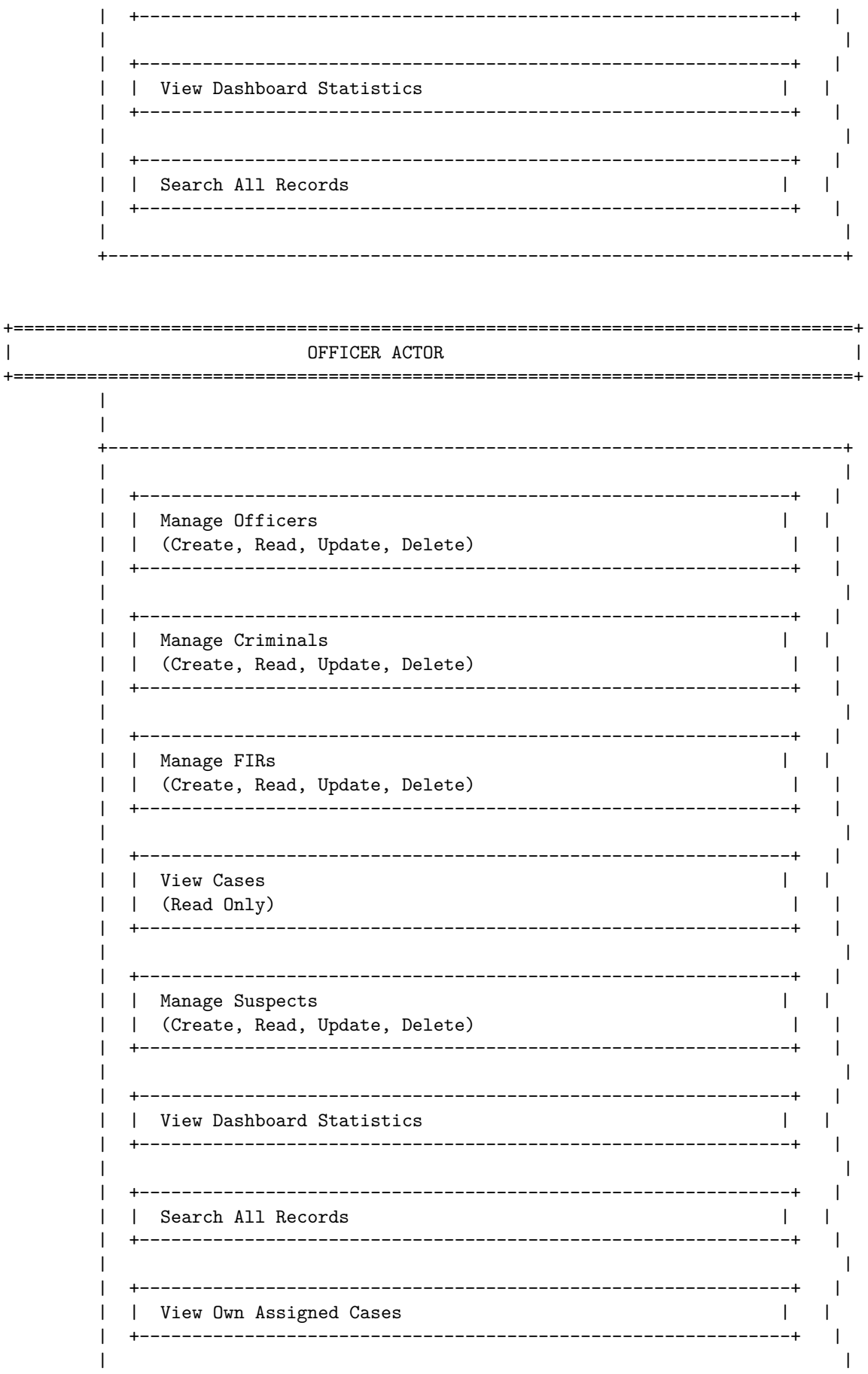
1. Use Case Diagram

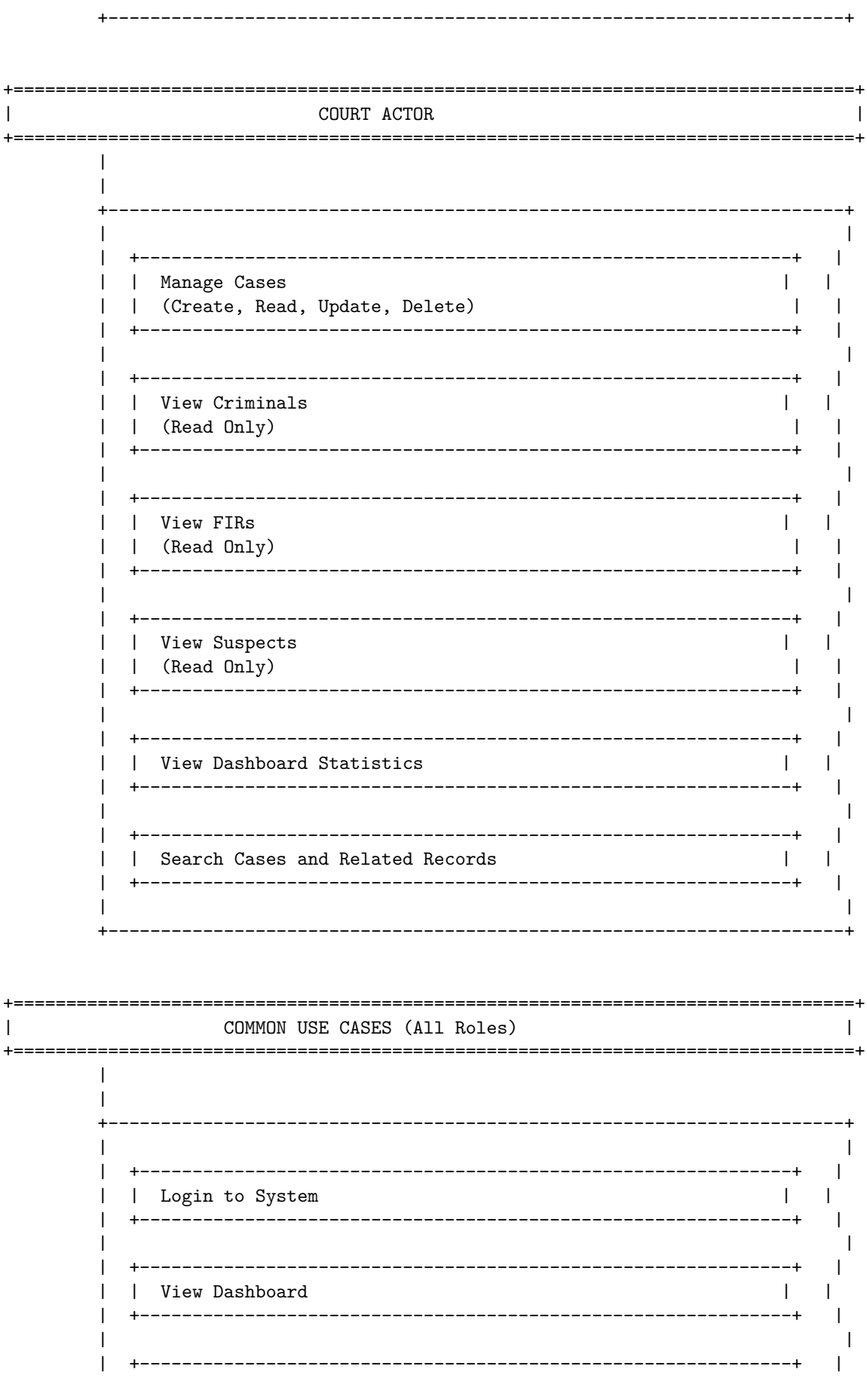
Actors

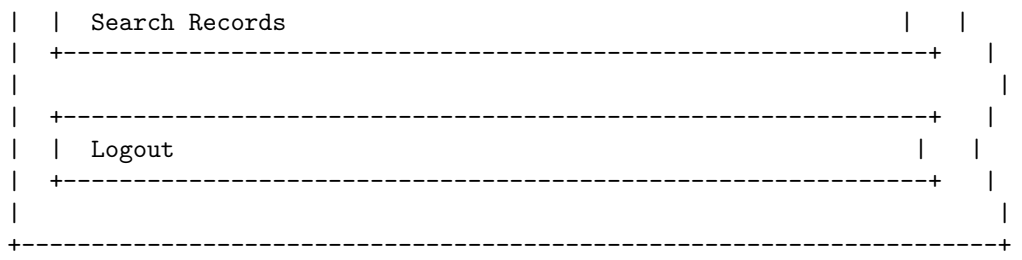
- **ADMIN**: System administrator with full access
- **OFFICER**: Police officer with data management access
- **COURT**: Court user with case management access

Use Case Diagram Structure



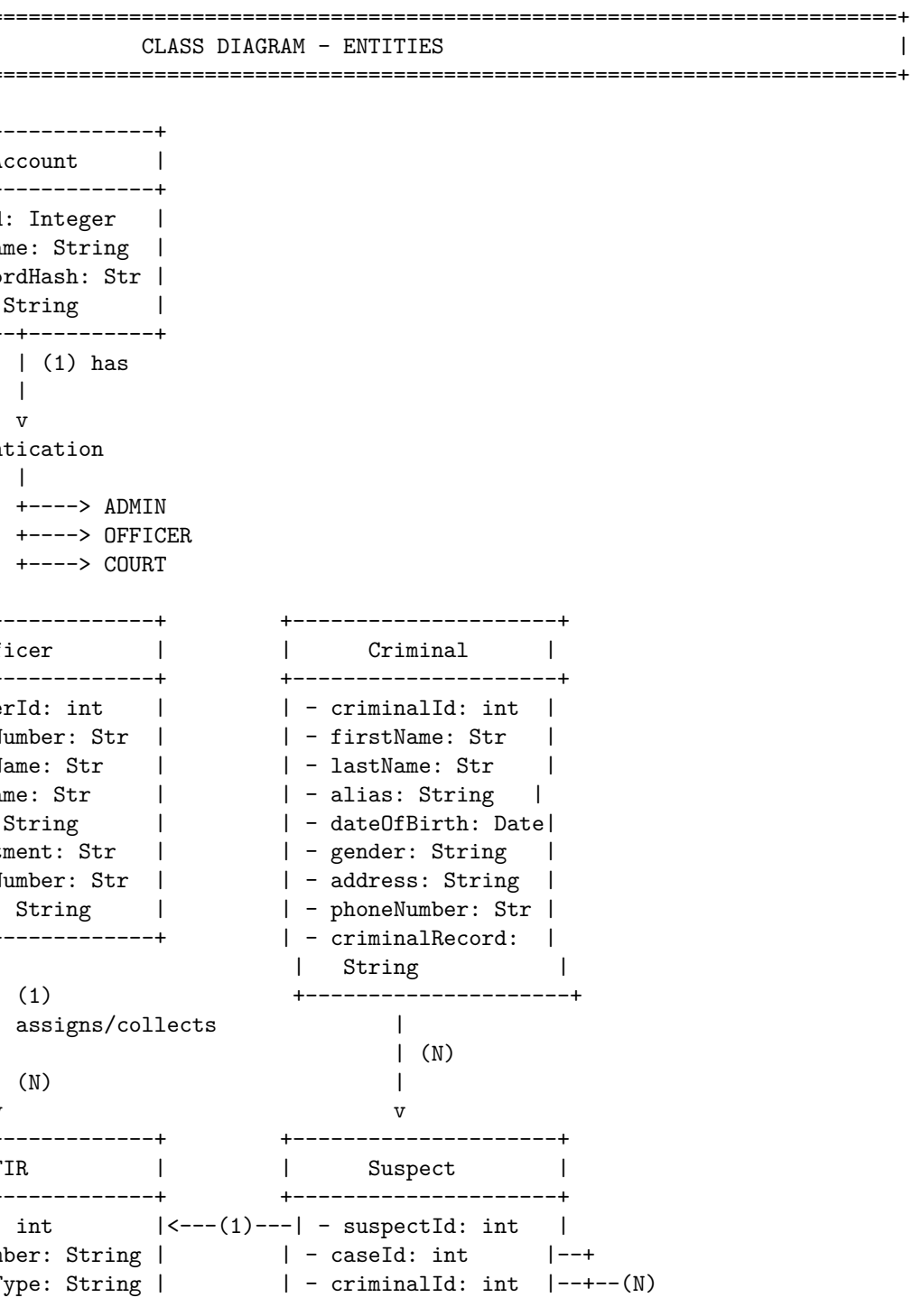

View Dashboard Statistics
Search All Records
OFFICER ACTOR
Manage Officers
(Create, Read, Update, Delete)
Manage Criminals
(Create, Read, Update, Delete)
Manage FIRs
(Create, Read, Update, Delete)
View Cases
(Read Only)
Manage Suspects
(Create, Read, Update, Delete)
View Dashboard Statistics
Search All Records
View Own Assigned Cases

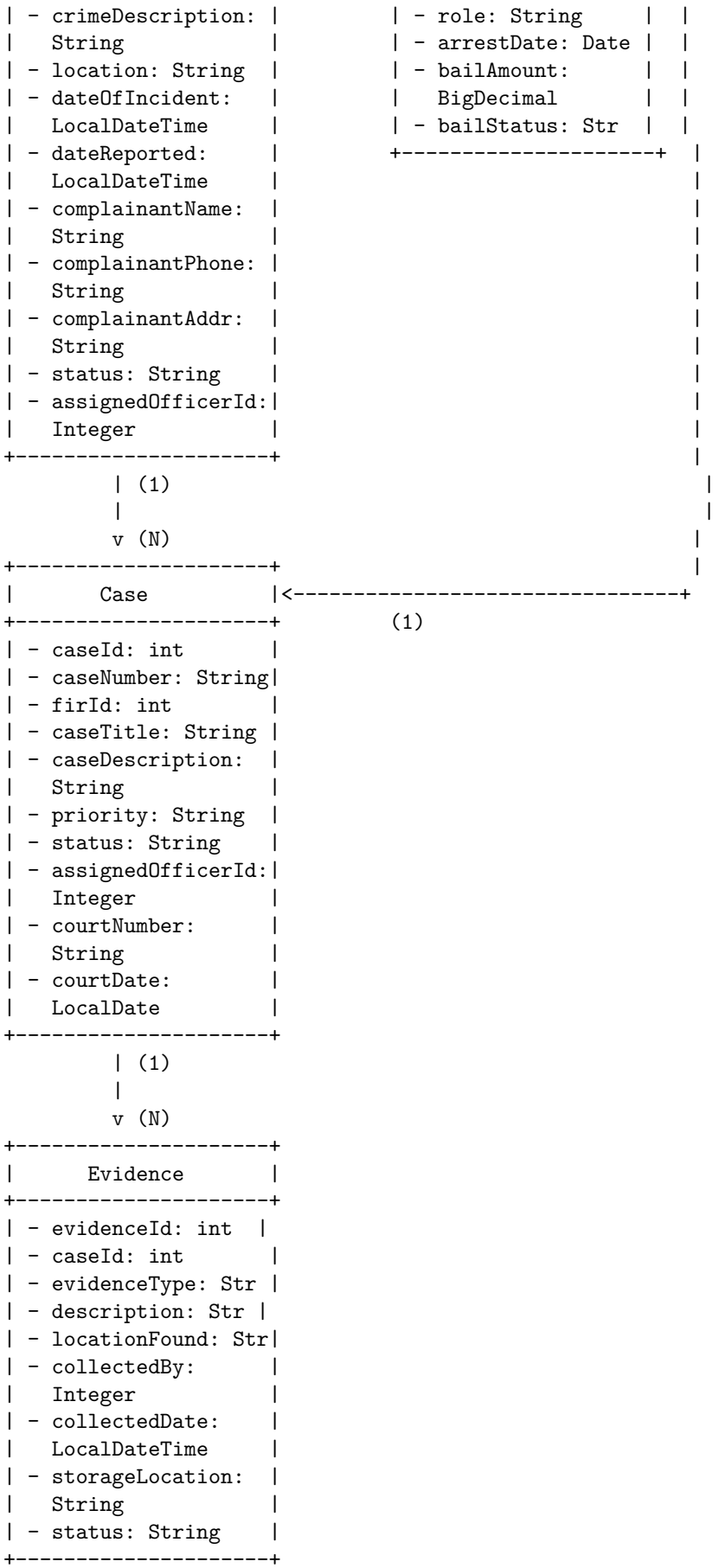

COURT ACTOR
Manage Cases
(Create, Read, Update, Delete)
View Criminals
(Read Only)
View FIRs
(Read Only)
View Suspects
(Read Only)
View Dashboard Statistics
Search Cases and Related Records
COMMON USE CASES (All Roles)
Login to System
View Dashboard

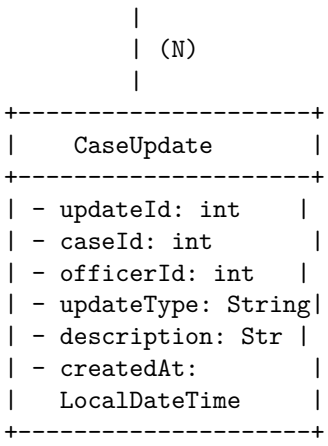


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2. Class Diagram - Entity Relationships




- crimeDescription: String
- location: String
- dateOfIncident: LocalDateTime
- dateReported: LocalDateTime
- complainantName: String
- complainantPhone: String
- complainantAddr: String
- status: String
- assignedOfficerId: Integer
- role: String
- arrestDate: Date
- bailAmount: BigDecimal
- bailStatus: Str
(1)
(N)
Case
(1)
- caseId: int
- caseNumber: String
- firId: int
- caseTitle: String
- caseDescription: String
- priority: String
- status: String
- assignedOfficerId: Integer
- courtNumber: String
- courtDate: LocalDate
(1)
(N)
Evidence
- evidenceId: int
- caseId: int
- evidenceType: Str
- description: Str
- locationFound: Str
- collectedBy: Integer
- collectedDate: LocalDateTime
- storageLocation: String
- status: String

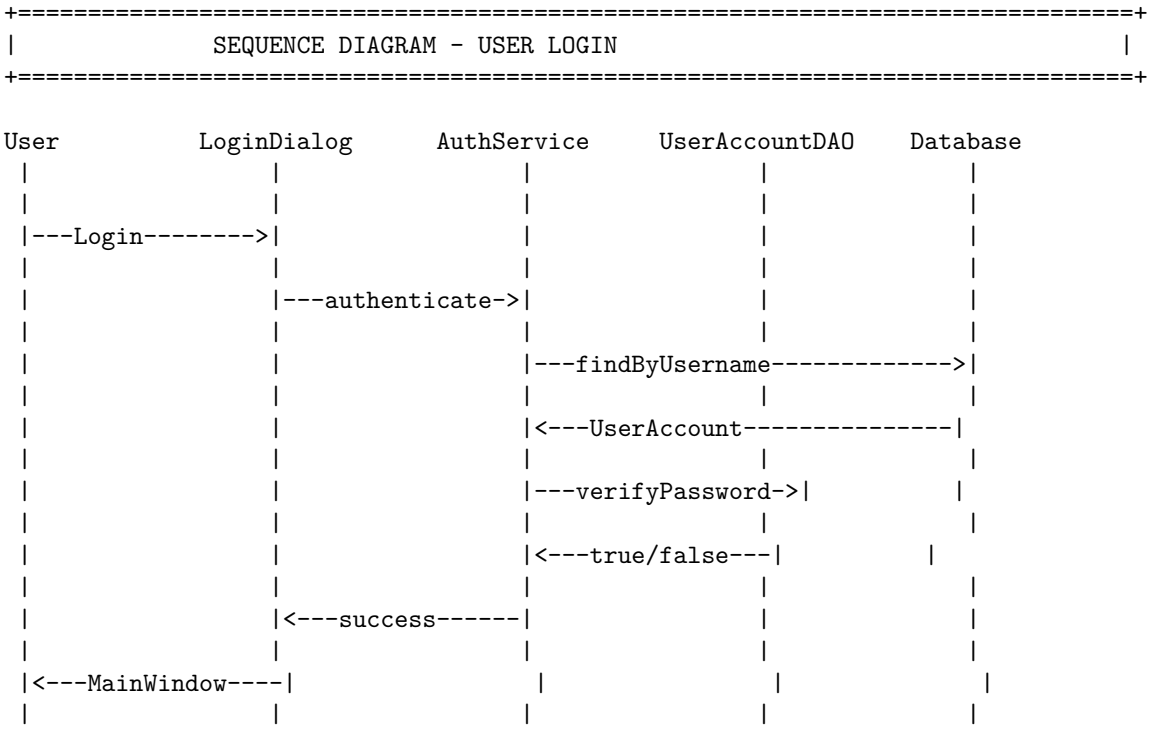


Relationship Summary:

- **UserAccount** → Authenticates users (ADMIN, OFFICER, COURT)
- **Officer** → Assigned to FIRs and Cases (1-to-Many)
- **FIR** → Linked to Cases (1-to-Many)
- **Case** → Linked to one FIR, many Suspects, Evidence, and CaseUpdates
- **Criminal** → Linked to Suspects (Many-to-Many via Suspect)
- **Suspect** → Links Case and Criminal (Many-to-Many relationship)
- **Evidence** → Belongs to Case (Many-to-One)
- **CaseUpdate** → Tracks Case history (Many-to-One)

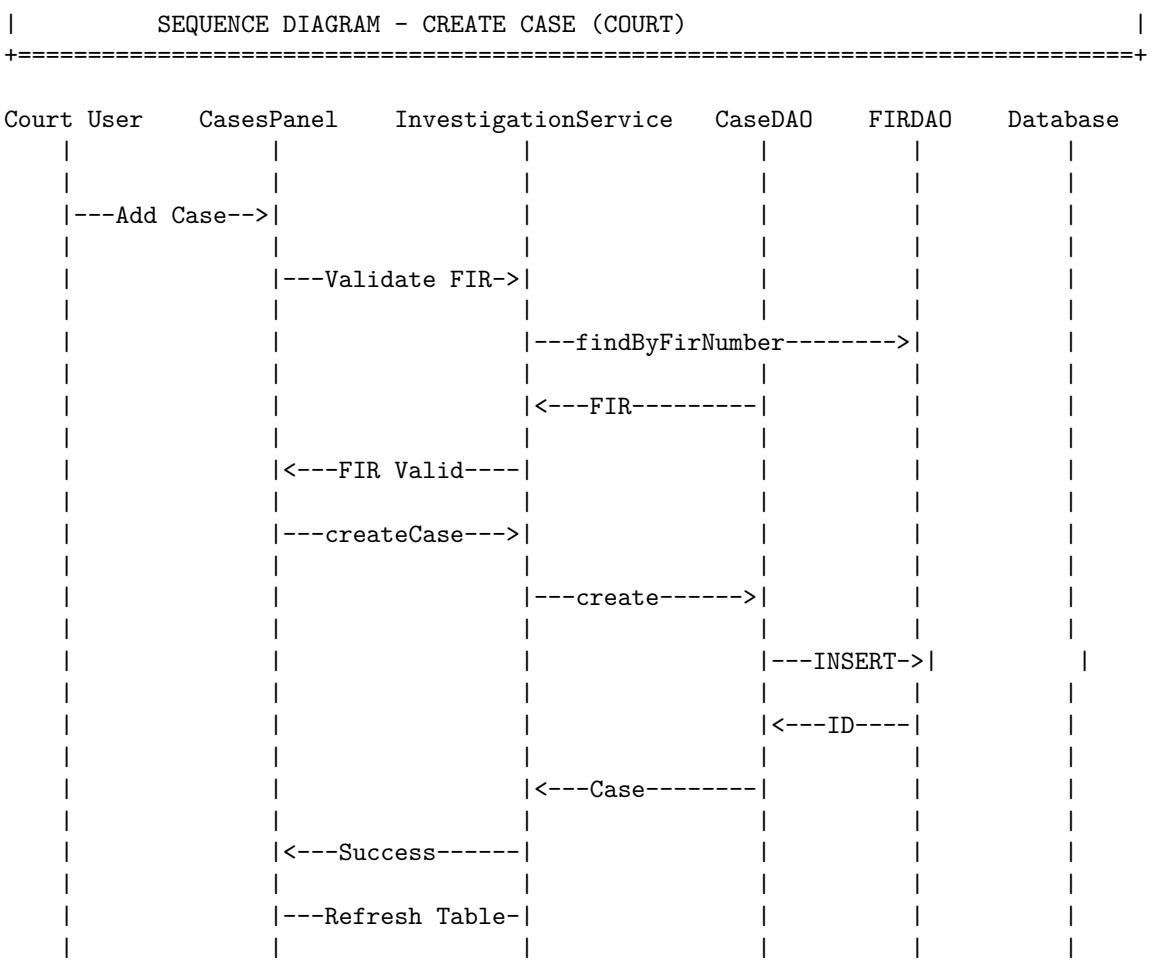
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3. Sequence Diagram - User Login Flow



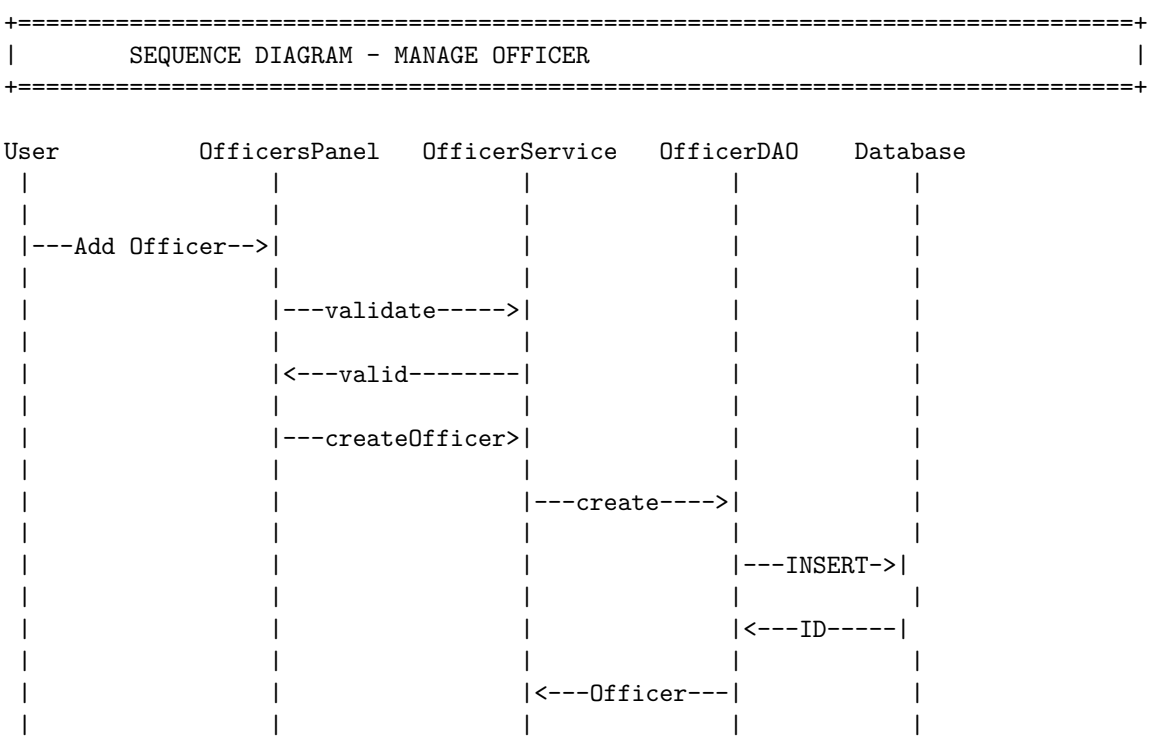
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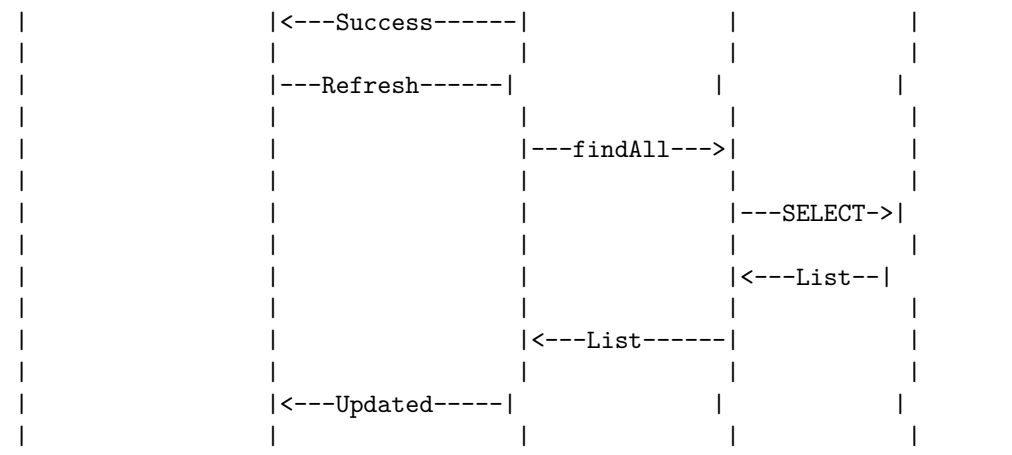
4. Sequence Diagram - Create Case Flow (COURT Role)



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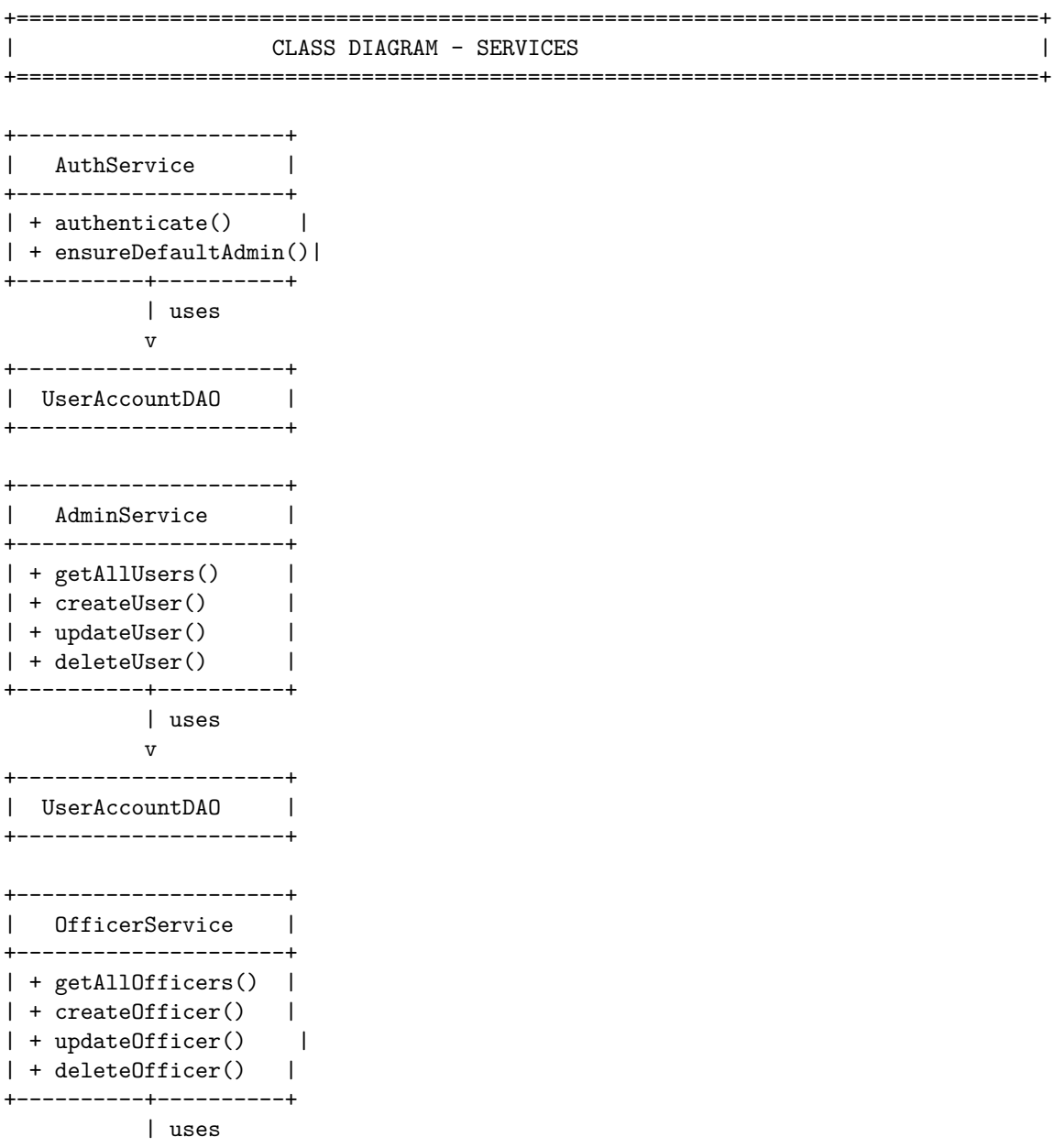
5. Sequence Diagram - Manage Officer Flow (OFFICER/ADMIN)

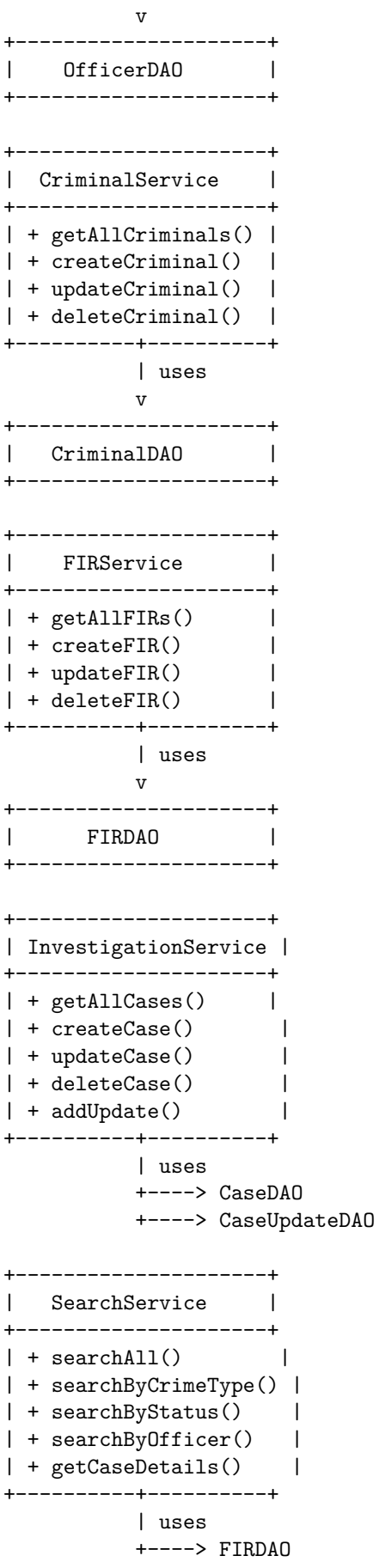




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6. Class Diagram - Service Layer




v
OfficerDAO
CriminalService
+ getAllCriminals()
+ createCriminal()
+ updateCriminal()
+ deleteCriminal()
uses
v
CriminalDAO
FIRService
+ getAllFIRs()
+ createFIR()
+ updateFIR()
+ deleteFIR()
uses
v
FIRDAO
InvestigationService
+ getAllCases()
+ createCase()
+ updateCase()
+ deleteCase()
+ addUpdate()
uses
+----> CaseDAO
+----> CaseUpdateDAO
SearchService
+ searchAll()
+ searchByCrimeType()
+ searchByStatus()
+ searchByOfficer()
+ getCaseDetails()
uses
+----> FIRDAO

```
           +----> CaseDAO
           +----> CriminalDAO
           +----> EvidenceDAO
           +----> OfficerDAO
           +----> SuspectDAO
           +----> CaseUpdateDAO
```

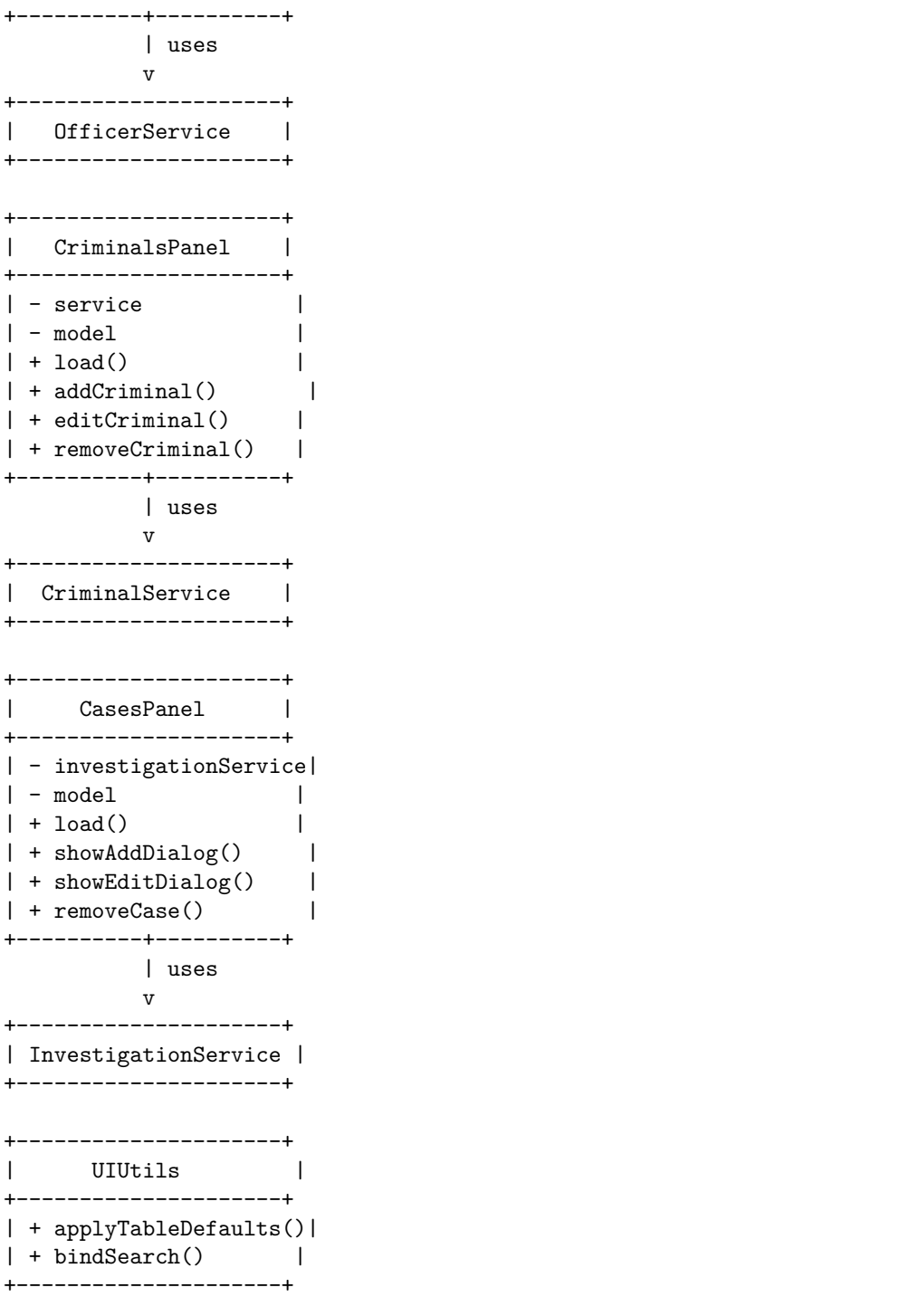
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7. Class Diagram - UI Layer

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+=============================================================================+
|                    CLASS DIAGRAM - UI LAYER                                 |
+=============================================================================+

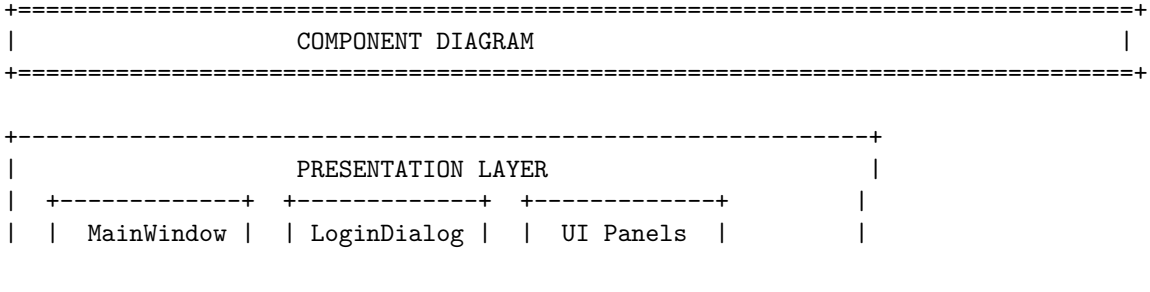
+---------------------+
|  DesktopLauncher    |
+---------------------+
| + main()            |
+----------+----------+
           | creates
           v
+---------------------+
|   LoginDialog       |
+---------------------+
| - username: JTextField|
| - password: JPasswordField|
| - role: JComboBox    |
| + getRole()          |
| + getUsername()      |
+----------+----------+
           | creates
           v
+---------------------+
|    MainWindow       |
+---------------------+
| - userRole: Role    |
| - username: String  |
| + buildMenu()        |
+----------+----------+
           | contains
           +----> StatsPanel
           +----> OfficersPanel (ADMIN/OFFICER)
           +----> CriminalsPanel
           +----> FIRsPanel
           +----> CasesPanel
           +----> SuspectsPanel
           +----> AdminPanel (ADMIN only)

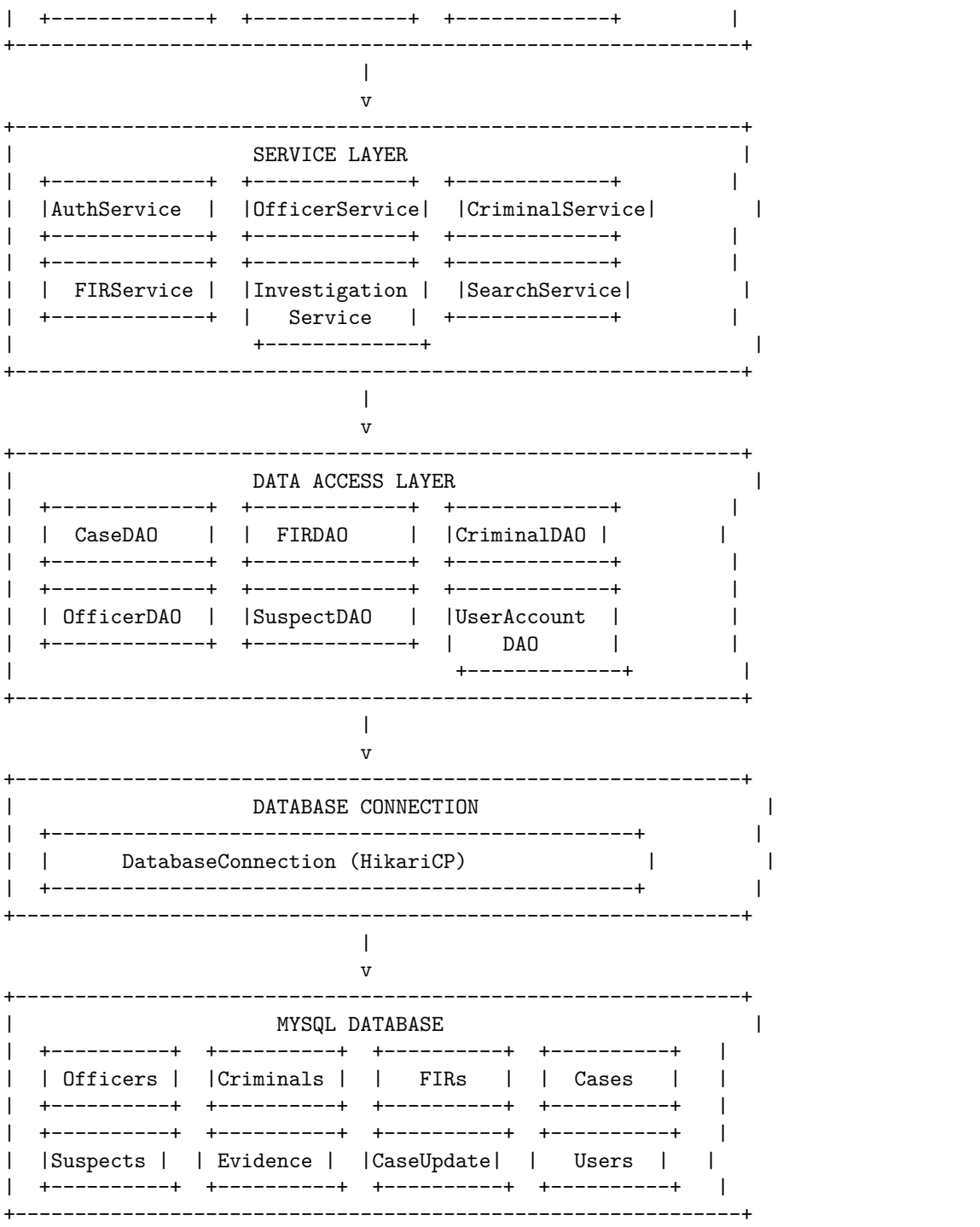
+---------------------+
|   OfficersPanel     |
+---------------------+
| - officerService     |
| - tableModel         |
| + loadOfficers()      |
| + onAdd()             |
| + onEdit()            |
| + onDelete()          |
```



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8. Component Diagram





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9. Activity Diagram - Case Creation Process

```
+=============================================================================+
|          ACTIVITY DIAGRAM - CREATE CASE                                     |
+=============================================================================+

[Start] --> [Court User clicks "Add Case"]
    |
    v
[Open Add Case Dialog]
```

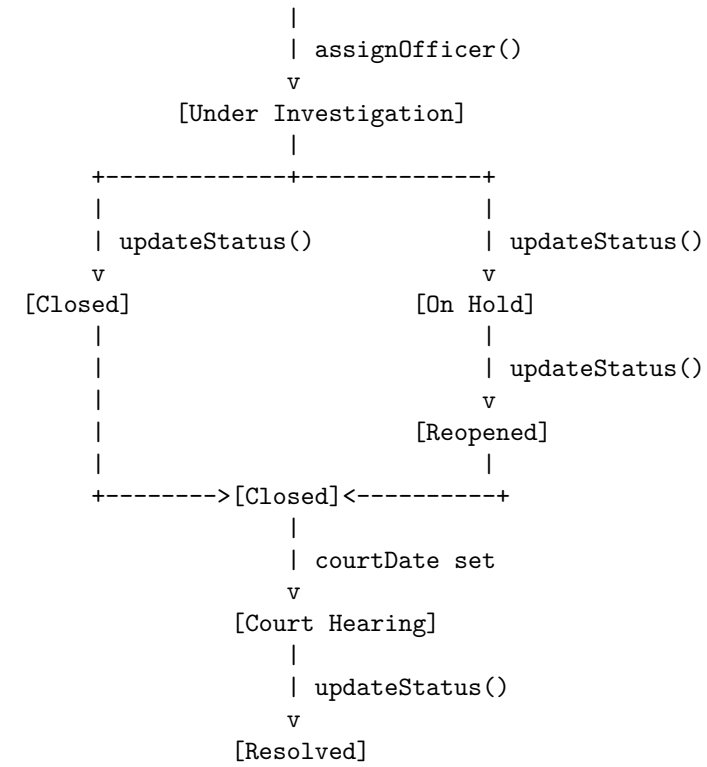
```
    |
    v
[Enter Case Details]
    |
    v
[Enter FIR Number/ID]
    |
    v
[Validate FIR Exists?]
    | No
    +--> [Show Error: FIR Not Found]
    +--> [Return to Dialog]
    | Yes
    v
[Validate Required Fields]
    | No
    +--> [Show Validation Error]
    +--> [Return to Dialog]
    | Yes
    v
[Validate Date Format]
    | No
    +--> [Show Date Error]
    +--> [Return to Dialog]
    | Yes
    v
[Create Case Object]
    |
    v
[Call InvestigationService.createCase()]
    |
    v
[Call CaseDAO.create()]
    |
    v
[Insert into Database]
    |
    v
[Case Created Successfully]
    |
    v
[Refresh Cases Table]
    |
    v
[Show Success Message]
    |
    v
[End]
```

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10. State Diagram - Case Status

```
+=============================================================================+
|              STATE DIAGRAM - CASE STATUS                                    |
+=============================================================================+

                    [Open]
```



Summary

Key Relationships:

1. **UserAccount** authenticates users with roles (ADMIN, OFFICER, COURT)
2. **Officer** can be assigned to multiple FIRs and Cases
3. **FIR** can have multiple Cases
4. **Case** belongs to one FIR, has many Suspects, Evidence, and Updates
5. **Suspect** links Cases and Criminals (many-to-many)
6. **Evidence** belongs to one Case
7. **CaseUpdate** tracks Case history

Role Permissions:

- **ADMIN**: Full system access including user management
- **OFFICER**: Manage officers, criminals, FIRs, suspects (read-only cases)
- **COURT**: Manage cases, read-only access to other data

Architecture Layers:

1. **Presentation**: Swing UI Panels
2. **Service**: Business logic layer
3. **DAO**: Data Access Object pattern
4. **Database**: MySQL with HikariCP connection pooling

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Tool: ASCII diagrams with square boxes